End-Term Sample Question Paper - SET A

Course Code: CSE563 Course Title: APPLIED MATHEMATICS

Maximum Marks: 60 Duration: 2 Hours Set A - all six units, theory and numerical

Academic Task Type: End-Term Sample Date: _____ Registration Number: _____

Instructions: answer all questions. Part A carries 2 marks each, Part B 5 marks each and Part C 10 marks each; in Part C the marks of each sub-part are shown in brackets. Show all working for numerical parts — an unsupported final answer earns no marks. The unit, course outcome and Bloom level are printed against every question.

Part A - Very Short Answer Questions (5 x 2 = 10 Marks)

Q1. State the division algorithm, and state the exact condition under which an integer a has a multiplicative inverse modulo m. **(2 Marks)**

[Unit I · CO1 · L1 - Remember · Theory]

Q2. For u = (1, −2, 2) and v = (3, 0, 4), compute u · v and ‖u‖. **(2 Marks)**

[Unit II · CO2 · L2 - Understand · Numerical]

Q3. Two fair dice are rolled. Find the probability that the sum is 10, listing the favourable outcomes. **(2 Marks)**

[Unit III · CO3 · L2 - Understand · Numerical]

Q4. State the handshaking lemma, and use it to find the number of edges in a graph whose degree sequence is (4, 3, 3, 2, 2, 2). **(2 Marks)**

[Unit IV · CO4 · L1 - Remember · Theory]

Q5. Define the notations O, Ω and Θ, and state what each one asserts about a running time. **(2 Marks)**

[Unit V · CO5 · L2 - Understand · Theory]

Part B - Short Answer Questions (4 x 5 = 20 Marks)

Q6. For the array [−3, 2, −1, 5, −4, 2], determine the maximum subarray sum and the subarray achieving it. Show the running best at each position, and state the running time of the scan. **(5 Marks)**

[Unit VI · CO6 · L3 - Apply · Numerical]

Q7. Use the Euclidean algorithm to find gcd(252, 105), then use the extended Euclidean algorithm to find integers x and y with 252x + 105y = gcd(252, 105). Verify your answer. **(5 Marks)**

[Unit I · CO1 · L3 - Apply · Numerical]

Q8. State Bayes' theorem, define prior, likelihood and posterior, and explain in two sentences why a highly accurate screening test can still produce mostly false alarms. **(5 Marks)**

[Unit III · CO3 · L2 - Understand · Theory]

Q9. Compare linear search and binary search on a sorted array of n elements: give the worst-case comparison count for each, state the precondition binary search needs, and state the lower bound for comparison-based sorting with a one-line reason. **(5 Marks)**

[Unit V · CO5 · L4 - Analyse · Theory]

Part C - Long Answer Questions (3 x 10 = 30 Marks)

Q10. Principal Component Analysis. **(10 Marks)**

[Unit II · CO2 · L4 - Analyse · Theory + Numerical]

(a) State the main steps of PCA and explain why the data must be centred before the covariance matrix is formed. *[5]*

(b) For the points (2,1), (3,2), (4,4), (5,3), (6,5): compute the mean, the 2×2 covariance matrix (divide by n − 1), both eigenvalues, and the percentage of variance carried by PC1. *[5]*

Q11. Shortest paths in a weighted directed graph. **(10 Marks)**

[Unit IV · CO4 · L4 - Analyse · Theory + Numerical]

(a) Describe the central idea of Dijkstra's algorithm, and explain why it can return a wrong answer when an edge weight is negative. *[5]*

(b) Apply it from A to the graph with edges A→B = 6, A→C = 2, C→B = 3, B→D = 1, C→D = 7, D→E = 4. Give the distance to every vertex, the finalisation order and each predecessor. *[5]*

Q12. Greedy and dynamic programming on the knapsack problem. **(10 Marks)**

[Unit VI · CO6 · L5 - Evaluate · Theory + Numerical]

(a) State the greedy-choice property and optimal substructure, and say which of the two knapsack versions has each. *[5]*

(b) Items are (2 kg, ₹20), (4 kg, ₹32) and (6 kg, ₹36) with capacity 10 kg. Solve the fractional version by the greedy ratio rule, give the true 0/1 optimum, and comment on the gap. *[5]*

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